

Why Discrete Price Fragments U.S. Stock Exchanges and Disperses Their Fee Structures

Yong Chao

University of Louisville

Chen Yao

Chinese University of Hong Kong

Mao Ye

University of Illinois, Urbana-Champaign and NBER

Stock exchange operators compete for order flow by setting “make” fees for limit orders and “take” fees for market orders. When traders can quote continuous prices, exchange operators compete on total fee, because traders can choose prices that perfectly neutralize any fee division. The 1-cent minimum tick size, however, prevents traders from neutralizing fee division. The nonneutrality of division between make and take fees (1) allows an exchange operator to establish exchanges that differ in fee structure to engage in second-degree price discrimination and (2) destroys the Bertrand equilibrium, leads to frequent fee changes, and encourages entries of new exchanges. (*JEL* G10, G20, L11, L13)

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All stock exchanges in the United States are for-profit entities. In most finance models, however, stock exchanges have no explicit role. How do

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Table 1
Structure of U.S. stock exchanges and the corresponding fee structures of each exchange

Holding companies	Exchange	Make fee	Take fee	Total fee
NASDAQ OMX Group	Philadelphia Stock Ex.	−\$0.0015	\$0.0025	\$0.0010
	Boston Stock Ex.	\$0.0020	−\$0.0004	\$0.0016
	NASDAQ Stock Market	−\$0.0020	\$0.0030	\$0.0010
Intercontinental Exchange	NYSE MKT	−\$0.0016	\$0.0028	\$0.0012
	NYSE ARCA	−\$0.0020	\$0.0030	\$0.0010
	New York Stock Ex.	−\$0.0015	\$0.0027	\$0.0012
Bats Global Markets	EDGX Exchange	−\$0.0020	\$0.0029	\$0.0009
	EDGA Exchange	\$0.0005	−\$0.0002	\$0.0003
	BYX Exchange	\$0.0018	−\$0.0015	\$0.0003
	BZX Exchange	−\$0.0020	\$0.0030	\$0.0010

This table depicts the ten U.S. stock exchanges run by three holding companies (operators). This table also displays the corresponding make/take fees on each exchange as of May 15, 2015. The table shows the standard fee structure for NYSE-listed stocks. For exchanges not stating standard rates, we present fees unconditional on the participants' activities.

stock exchanges set their service fees? How does fee competition shape the organization of the stock exchange industry? In this paper, we propose a model to examine the role of discrete pricing in creating two salient features of the U.S. stock exchange industry: fee dispersion and market fragmentation.

All stock exchanges in the United States are organized as electronic limit order markets and a stock can be traded on any of them.¹ A trader can act as a liquidity maker by posting a limit order. A trade occurs when another trader (a liquidity taker) accepts the terms of a limit order through a market order. The exchanges charge a “make” fee and a “take” fee per share to each side of a transaction, the sum of which, the so-called “total” fee, is a major source of exchanges' revenues (O'Donoghue 2015). Table 1 shows a significant variation in fees across competing exchanges, as well as across exchanges owned by the same holding company (hereafter “operator”). Also, “the pressure to establish novel and competitive pricing often leads exchanges to modify their pricing frequently, typically on a calendar-month basis” (U.S. Security and Exchanges Commission (SEC) 2015a, p. 21). Such spatial and temporal dispersion of prices cannot be justified by physical product differentiation, as the exchanges are so similar that the SEC even refers to some of them as “cloned markets” (SEC 2015a, p. 22).

According to the principle of tax neutrality, at a given tax level, it does not matter who—buyer or seller—is liable for the tax. Therefore, all traders would choose the exchange with the lowest total fee. Competition over the total fee should lead to a Bertrand equilibrium, resulting in zero profit, and leaving no room for market entry with any fixed cost to establish an exchange. Yet Table 1 demonstrates two types of market fragmentations: (1) multiple operators coexist and (2) each operator offers multiple stock exchanges. New market entries are

¹ Unlisted trading privileges in the United States allow stocks to be traded outside the listing venue.

commonly observed. For example, on October 22, 2010, Bats created a new stock exchange, Bats Y, in addition to its existing Bats X exchange.

We show that one driving force behind fee dispersion and market fragmentation is the discrete tick size.² When liquidity makers quote continuous prices, the breakdown of make/take fees is inconsequential (Colliard and Foucault 2012). Liquidity makers always choose the exchange with the lowest total fee, and they neutralize the make/take fee allocations by adjusting their quotes. Thus, operators compete only on total fees. No operator has incentives to offer multiple exchanges, and their competition over the total fee leads to a Bertrand equilibrium. When liquidity makers have to quote discrete price, they cannot neutralize fee breakdowns (Foucault, Kadan, and Kandel 2013). Thus, operators compete on both make and take fees.

We find that the nonneutrality of fee breakdowns creates vertical product differentiation for otherwise identical exchanges. All else being equal, a liquidity maker prefers the exchange with a lower take fee, because it increases the probability that a liquidity taker will accept the offer. An exchange with a lower take fee, therefore, has higher quality. A monopoly operator can then create exchanges with differentiated *prices* (the make fees) and *qualities* (the take fees) to screen liquidity makers. Liquidity makers with larger (smaller) gains from execution select the exchanges with higher (lower) price and higher (lower) quality. We show that such second-degree price discrimination increases the operator's profit, as well as the welfare of both liquidity makers and liquidity takers, because new exchanges create more effective transaction prices.

We then show that no pure strategy equilibrium exists with duopoly operators, each opening one exchange. Similar to the Bertrand equilibrium, a positive total fee leads to incentives to undercut toward a zero total fee. Surprisingly, two types of profitable deviations rule out any pure strategy equilibrium with zero total fees. One type increases price (make fee) and decreases the quality (take fee), which attracts liquidity makers with high gains from execution. The other type decreases price and increases quality, which appeals to liquidity makers with low gains from execution. The nonexistence of a pure-strategy equilibrium provides one rationalization for the frequent fee changes. We then show that any mixed-strategy equilibrium generates positive profits, which provides a justification for the entry of new exchanges.

Finally, we find that the nature of the fee game is to restore continuous prices. Each liquidity maker has an ideal limit order price that balances the gains from execution and execution probability. In our model, the valuations of liquidity makers are continuous, as are their ideal limit order prices. The discrete tick size forces almost all liquidity makers to quote suboptimal limit order prices.

² SEC Rule 612(c) of Regulation National Market Systems (Reg. NMS) imposes a uniform tick size (minimum price variation) of one cent for securities priced equal to or greater than \$1.00 per share. For other studies on tick size, see Kandel and Marx (1997), Anshuman and Kalay (1998), Buti et al. (2015), Yao and Ye (2018), and Wang and Ye (2017), among others. For a history of tick size, see Harris (2003).

With free entry, our game reaches pure strategy equilibria only when exchanges' fee structures span continuous price. If not, an operator can create a new fee structure to attract liquidity makers. The free entry equilibria involve infinitely many stock exchanges with zero total fee. In reality, we cannot satisfy such conditions for pure-strategy equilibria because the number of exchanges is finite, which explains the diversity and frequent fee changes and new exchanges. The free entry equilibria implies that fee competition can be considered as exchange operators' financial innovations to restore continuous prices, which is complementary to financial innovations that span all states of nature in an incomplete market (Duffie and Rahi 1995; Grinblatt and Longstaff 2000).

Our tick size channel provides the first interpretation for market fragmentation within the same operator. (Foucault, 2012, p. 20) conjectures that an operator establishes exchanges with heterogeneous fees to screen investors by type. Nevertheless, "it is not clear however how the differentiation of make/take fees suffices to screen different types of investors" (Foucault 2012, p. 20). In this paper, we show that the nonneutrality of the fees allows operators to screen liquidity makers based on the cumulative (cum) fee prices to liquidity takers.

We also provide a new interpretation of market fragmentation across operators. Models with coexisting exchanges usually involve physical production differentiation such as matching rules (Gehrig 1993; Parlour and Seppi 2003), transparency (Madhavan 1995; Barclay, Hendershott, and McCormick 2003), margin requirements (Santos and Scheinkman 2001), and speed (Pagnotta and Philippon 2016). However, the SEC (2015a) points out the major stock exchanges are almost identical. In addition, U.S. regulations and technological improvements have gradually reduced a number of frictions that could fragment the market, such as search cost (Spulber 1996a, 1996b) and market impact (Pagano 1989; Ellison and Fudenberg 2003; Malamud and Rostek 2017). Foucault and Menkveld (2008) find that exogenous exchanges can coexist with tick size constraints. We contribute to the literature by showing that tick size constraints allow physically identical exchanges to coexist when they compete on fees.

We are the first to rationalize exchange fee dispersion and frequent fee changes, documented by Cardella, Hao, and Kalcheva (2015) and O'Donoghue (2015). Colliard and Foucault (2012) predict that competition between exchanges leads to zero total fees. Cardella, Hao, and Kalcheva (2015) and Malinova and Park (2015) maintain that rebates to liquidity makers encourage liquidity provision, which, in turn, attracts liquidity takers. Yet this interpretation does not clarify why new market entrants, such as Direct Edge A and Bats Y, charge liquidity makers positive fees. Angel, Harris, and Spatt (2010, 2013) and Battalio, Corwin, and Jennings (2016) find that retail brokers have incentives to route customer orders to exchanges with the highest rebates, because brokers do not need to pass the rebates along to their customers. But this interpretation does not make clear why exchanges do not simply copy the highest rebate set by their competitors.

On October 3, 2016, the SEC implemented a pilot program to increase the tick size from 1 cent to 5 cents for 1,200 small stocks. We predict that an increase in tick size tends to increase the dispersion of the fee structure. Because SEC Rule 610 imposes a cap of 0.3 cents per share for a take fee, it would be interesting to see whether an increase in tick size for less frequently traded stocks would induce exchanges to charge higher make fees. On June 8, 2016, the Equity Market Structure Advisory Committee of the SEC recommended a pilot program to reduce the cap of the take fee. We predict that this reduction would also increase the make fee, but not the dispersion of the fees. To isolate the impact of each pilot program, we encourage the SEC to create the following 2-by-2 pilot groups: (1) change the tick size and take fee cap; (2) change only the tick size; (3) change only the take fee cap; and (4) no change.

Our model also informs the literature on two-sided markets, vertical product differentiation, and price dispersion. Rochet and Tirole (2006, p. 646) define a market as two-sided if “the volume of transactions between end users depends on the structure and not only on the overall level of the fees charged by the platform.” A fundamental question is whether this two-sidedness can generate qualitatively different predictions from one-sided markets with identical setups.³ We show that nonneutrality creates product differentiation among intrinsically homogeneous exchanges, which leads to fee dispersion and market fragmentation. For games with vertical production differentiation, Shaked and Sutton (1982) and Tirole (1988) find pure-strategy equilibrium when quality is chosen before price. In our model, the “qualities” of the exchanges for liquidity makers are determined by the take fees, which can be adjusted as easily as the make fees. Such simultaneous choices of price and quality destroy any pure-strategy equilibrium. Researchers use the mixed-strategy equilibrium to rationalize spatial price dispersion (different prices at the same time) and the temporal price dispersion (price fluctuations over time). To prevent all consumers from flocking to the market with the lowest price, the literature on one-dimensional price competition assumes frictions for consumers to obtain the prices (Rosenthal 1980; Varian 1980; Burdett and Judd 1983) or assumes frictions for firms to advertise their prices (Butters 1977; Baye and Morgan 2001). Our model does not include such market frictions, and a pure-strategy equilibrium does not exist, because prices cannot be uniquely ranked in a two-dimensional space.

1. Institutional Details and Literature Review

The SEC (2015b) considers Regulation NMS as the main force to fragment stock trading. Regulation NMS has five parts, three of which (Rule 610, Rule 611, and Rule 612) relate to market fragmentation.

³ For alternative definitions of two-sided markets, see Rysman (2009), Weyl (2010), and Hagiu and Wright (2015).

The Order Protection Rule (SEC Rule 611) requires stock exchanges to route incoming market orders to the market posting the best nominal quote. This Rule rules out market fragmentation due to geographic location (Cantillon and Yin 2011). The market impact model by Pagano (1989), Ellison and Fudenberg (2003), and Malamud and Rostek (2017) shows that multiple exchanges can coexist if each of them has an independent price, but Rule 611 links the best nominal price across exchanges. Spulber (1996a), 1996b) shows that an increase in search cost softens the price competition of intermediaries and encourages market fragmentation. The Rule 611, however, reduces search costs, a move that should consolidate the market.

Rule 611 also reduces physical product differentiation across U.S. stock exchanges. After Regulation NMS, U.S. stock exchanges converged to the same market structure: the electronic limit order book (SEC 2015a), because Regulation NMS only protects electronic quotes.⁴ This convergence rules out the interpretation of fragmentation based on differentiated trading rules (Gehrig 1993; Hendershott and Mendelson 2000; Parlour and Seppi 2003; O'Hara and Ye 2011; Zhu 2014; Ye 2015). To prevent exchanges from competing on the coarseness of their grid size, Rule 612 of Regulation NMS imposes a uniform tick size (Biais, Bisière, and Spatt 2010), effectively eliminating product differentiation on minimum price variation.

Rule 610 of Regulation NMS, the access rule, prevents exchanges from giving preferential treatment to members and/or subscribers over nonmembers and/or nonsubscribers. Cantillon and Yin (2011) find that large switching costs fragment the market. Rule 610 reduces switching costs and thus should consolidate trading.

Because the Rule 611 applies to the nominal price but not the cum fee price, an exchange has incentives to manipulate nominal price by increasing rebates to liquidity makers. Rule 610 imposes a maximum take fee of 0.3 cents to limit the size of the rebates an exchange can offer to liquidity makers. Because brokers do not need to pass these rebates along to their customers, Angel, Harris, and Spatt (2010, 2013) and Battalio, Corwin, and Jennings (2016) find that brokers tend to route customer limit orders to exchanges with the highest rebates. Despite these distorted incentives to offer rebates to limit orders, two recent entrants into the stock exchange industry, Direct EDGE X and Bats Y, choose to charge liquidity makers. Our model provides an interpretation for why some exchanges choose to charge liquidity makers.

Pagnotta and Philippon (2016) show that Regulation NMS can fragment the market if the speed of exchanges is different. In this paper, we explain why operators establish multiple exchanges with similar speeds, such as Direct EDGE A and Direct EDGE X. Spatt (2016) argues that a market fragments because Rule 611 only protects the best quotes in each exchange, which

⁴ Slow manual quotes may grind trading to a halt in the rerouting process (Hasbrouck 2007).

promotes the proliferations of exchanges compared with the case under a more aggressive alternative that protects the quotes all the way down the book.

After ruling out these possible explanations of market fragmentation and fee dispersion across “cloned exchanges,” we show how discrete price fragments the markets and disperses their fees. Certainly, the profit margin from the fees is small. The move to electronic trading, partially driven by Regulation NMS, also contributes to market fragmentation and fee dispersion, as it reduces the costs to establish exchanges and collect fees.

To change its fee structure, an exchange needs to file a rule change with the SEC, which publishes it for public comment. Although fee filings are not subject to SEC approval, the SEC may, within 60 days of the filing, suspend the new fee and institute proceedings to determine whether or not to approve it (SEC 2015a, p. 4.).

2. Model

We consider a three-period game among three types of risk-neutral players: exchange operator(s), a continuum of liquidity makers with valuation of a stock v_b uniformly distributed on $[\frac{1}{2}, 1]$, and liquidity takers with valuation of stock v_s uniformly distributed on $[0, \frac{1}{2}]$. v_b and v_s , respectively, are the liquidity makers’ and liquidity takers’ private information. Because liquidity makers have higher valuations, they intend to buy from the liquidity takers. The Online Appendix shows that our predictions do not change when liquidity makers sell to liquidity takers.

In the game, we consider both a monopoly exchange operator and competition between exchange operators. Figure 1 shows the actions available to players in each period. At date 0, operators simultaneously choose the number of exchanges to establish and set the make/take fees to maximize their expected profits. We use (f_m^i, f_t^i) to label these fee structures, unless we consider multiple operators who own multiple exchanges each. Fees are charged upon order execution, with a negative fee meaning a rebate for the trader.

At date 1, nature randomly draws one liquidity maker from the uniform distribution $[\frac{1}{2}, 1]$. The liquidity maker makes two decisions after observing the fee structures: (1) submitting a limit order of one share to exchange i or to no exchange at all and (2) proposing the limit order at price P^i if choosing exchange i . The liquidity maker aims to maximize the expected surplus:

$$BS^i(P^i; v_b) \equiv (v_b - P^i - f_m^i) \cdot \Pr(v_s \leq P^i - f_t^i).$$

$(v_b - P^i - f_m^i)$ is the liquidity maker’s gains from execution, and $\Pr(v_s \leq P^i - f_t^i)$ reflects the probability of the liquidity taker accepting the limit order. We assume that the liquidity maker cannot submit a limit order

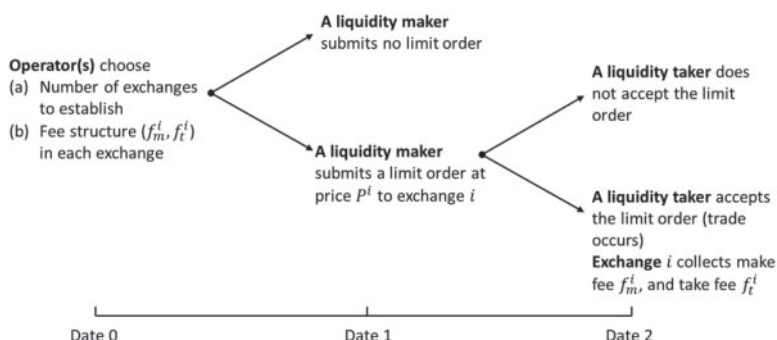


Figure 1
Time line of the model

price so low that no liquidity taker will accept it.⁵ The liquidity maker submits no limit order if the maximal expected surplus is negative.

At date 2, nature randomly draws one liquidity taker from the uniform distribution $[0, \frac{1}{2}]$. The liquidity taker chooses to accept or decline the limit order, accepting the limit order if the surplus is nonnegative after paying the limit order price and the take fee. The exchange where the trade occurs profits from the total fee.⁶

We solve the subgame perfect equilibrium of the sequential-move game through backward induction. First, we look for the probability that a liquidity taker accepts the limit order at price P^i given the take fee f_t^i , $\Pr(v_s \leq P^i - f_t^i)$. Then we study the liquidity maker's exchange choice and the optimal limit order price P^{i*} to maximize the expected surplus, given (f_t^i, f_m^i) and the liquidity taker's best response at date 2. The liquidity maker chooses not to participate if the expected surplus is negative. Finally, given the probability of participation based on the above subgames, a monopoly operator chooses the number of exchanges and the associated fee structures to maximize its expected profit, while competing operators choose the number of exchanges and the associated fee structures simultaneously, such that those strategies form a Nash equilibrium.

As a model on exchange competition, our model is parsimonious with traders' choices between limit and market orders: they do not choose the order type and the limit order book is empty when the liquidity maker arrives.⁷ This

⁵ This type of stub quotes can mechanically lead to zero expected surplus. The SEC, however, prohibits market maker stub quotes. See <https://www.sec.gov/news/press/2010/2010-216.htm>.

⁶ In practice, an exchange charges a fee to route a market order to another exchange. Doing so imposes a barrier to submitting an order to a low take fee exchange aiming to interact with a limit order in a high take fee exchange for to reduce the trading cost. For an example of a routing fee, see https://www.batstrading.com/support/fee_schedule/edga/.

⁷ In practice, the decision between making and taking liquidity is not so rigid. Yet Menkveld (2013) and Foucault, Kadan, and Kandel (2013) maintain that high-frequency traders account for a large fraction of liquidity supply in electronic markets and that specialization among traders on making and taking exists.

arrangement allows us to extend the model's economic insights beyond the stock exchange industry (see the Conclusion).

Under zero tick size, P^i can be any real number. Under a discrete tick size, P^i can only be on price grids with the minimum distance as tick size; we call such restrictions tick size constraints. Our main analysis in Sections 4–7 focuses on a tick size of 1, that is,

$$P^i \in \{\dots, -1, 0, 1, \dots\}. \quad (1)$$

In Equation (1), the valuations of the liquidity maker and taker fall within the same tick. Section 8 shows that our results still hold when the support of traders' valuation goes beyond one tick.

3. Tick Size Constraints Lead to Nonneutrality of Fees

Similar to Foucault, Kadan, and Kandel (2013), tick size constraints create nonneutrality of fees in our model. We derive the nonneutrality result in our framework to provide a foundation to discuss our new findings in Sections 4–7. In Subsection 3.1, we show that fee breakdowns are neutral under continuous pricing. In Subsection 3.2, we show how discrete price destroys the neutrality of fee breakdowns.

3.1 Fee neutrality under continuous pricing

In this section, P^i can be any real number. At date 2, a liquidity taker accepts a limit order if $v_s \leq P^i - f_t^i$, and the probability for the liquidity taker to accept the limit order is $\Pr(v_s \leq P^i - f_t^i)$. At date 1, a liquidity maker proposes a limit order price to maximize the expected surplus, $BS^i(P^i; v_b)$:

$$\max_{P^i} BS^i(P^i; v_b, f_m^i, f_t^i) = \max_{P^i} [(v_b - P^i - f_m^i) \cdot \max\{0, \min\{2 \cdot (P^i - f_t^i), 1\}\}]. \quad (2)$$

We denote the total fee on exchange i as $T^i \equiv f_m^i + f_t^i$ and its cum fee buy price as $p_b^i \equiv P^i + f_m^i$. The liquidity maker's optimization problem becomes

$$\max_{p_b^i} BS^i(p_b^i; v_b, T^i) = \max_{p_b^i} [(v_b - p_b^i) \cdot \max\{0, \min\{2 \cdot (p_b^i - T^i), 1\}\}]. \quad (2')$$

Equation (2') shows, under a zero tick size, a liquidity maker's choice depends only on the total fee T^i , but not its breakdown into f_m^i and f_t^i . This neutrality result holds for any distribution of v_b . The uniform distribution yields an analytical solution for equations (2) and (2'):

$$P^{i*}(v_b) = \frac{v_b + T^i}{2} - f_m^i \text{ if } v_b \geq T^i, \text{ and} \quad (3)$$

$$p_b^{i*}(v_b) = \frac{v_b + T^i}{2}, \text{ if } v_b \geq T^i. \quad (3')$$

The cum fee sell price is

$$p_s^{i*}(v_b) = P^{i*} - f_t^i = \frac{v_b - T^i}{2}, \text{ if } v_b \geq T^i. \quad (4)$$

Equations (3), (3'), and (4) confirm the neutrality result established by Colliard and Foucault (2012). Holding the total fee fixed, if an exchange increases the make fee and decreases the take fee, a liquidity maker would reduce the nominal price P^i by the same amount, leaving the cum fee buy, cum fee sell, and the economic outcomes unchanged.

The maximal surplus for a buyer conditional on choosing exchange i is

$$BS^{i*}(v_b) = (v_b - P^{i*}(v_b) - f_m^i) \cdot \Pr(v_s \leq P^{i*}(v_b) - f_t^i) = \frac{(v_b - T^i)^2}{2}, \text{ if } v_b \geq T^i. \quad (5)$$

All traders then choose the exchange with the lowest total fee. As a result, a monopoly operator has no incentive to open more than one exchange, and competition among operators leads to a Bertrand equilibrium. In Proposition 1, we summarize the market outcomes under zero tick size.

Proposition 1 (Market outcome under zero tick size). Under continuous pricing,

- a only the total fee, not its breakdown, affects the market outcome;
- b a monopoly operator has no incentive to open more than one exchange.
The standing exchange charges a total fee of $\frac{3}{8}$ and earns a profit of $\frac{9}{64}$; and
- c competing operators set zero total fees and earn zero profits.

Proposition 1 implies no dispersion of total fees if the tick size is zero. As long as establishing an exchange involves nonzero costs, trading should consolidate to one operator and one exchange. Nevertheless, these predictions are not consistent with the stylized facts.

3.2 Nonneutrality of fee structures under tick size constraints

Next, we show that tick size constraints destroy fee neutrality. To simplify notation, we show the nonneutrality result using one exchange. Three necessary conditions for a trade to occur are (a) the seller's participation: $0 \leq v_s \leq P - f_t$; (b) the buyer's participation: $P + f_m \leq v_b \leq 1$; and (c) the exchange's participation: $0 \leq f_m + f_t$, which is equivalent to $P - f_t \leq P + f_m$. Combining conditions (a)–(c) leads to

$$0 \leq v_s \leq P - f_t \leq P + f_m \leq v_b \leq 1. \quad (6)$$

Lemma 1 directly follows from Equation (6).

Lemma 1 (Necessary conditions for positive trading volume). Under tick size constraints in Equation (1), in order for a trade to happen and the exchange to survive, the exchange cannot charge both sides. Moreover, the total fee cannot exceed the tick size:

$$f_m \cdot f_t \leq 0 \text{ and } 0 \leq f_m + f_t < 1. \quad (7)$$

Rochet and Tirole (2006, p. 646) define a market as two-sided if “the volume of transactions between end-users depends on the structure and not only on the overall level of the fees charged by the platform.” Lemma 1 shows that fee breakdowns affect trading volume when the tick size is one. For example, consider a monopoly exchange with a zero total fee. Trading volume is zero when $f_m = f_t = 0$: liquidity makers always quote a nominal buy price of zero, which is rejected by all liquidity takers, except for the knife-edge cases of $v_b = 1$ or $v_s = 0$. Trading volume is one when $f_m = 0.5$ and $f_t = -0.5$: liquidity makers always propose a nominal buy price of zero, which is always accepted by liquidity takers due to the rebates. Compared with Proposition 1, Lemma 1 shows that tick size constraints transform a stock exchange from a one-sided market into a two-sided market.

Equation (7) shows that, to achieve positive trading volume, an exchange operator cannot charge both sides. This result no longer holds after we reduce the tick size in Section 8, yet the prediction is consistent with stylized facts. Among the 108 fee structure changes in U.S. stock exchanges documented by Cardella, Hao, and Kalcheva (2015), exchanges differ on which side they subsidize, but no exchange ever charges both sides positive fees.⁸

Lemma 2 demonstrates the fee breakdowns that achieve the same cum fee buy/sell prices.

Lemma 2 (Equivalent fee breakdowns with one exchange). Under tick size constraints in Equation (1):

- a. If $0 \leq f_m < 1$, then $-f_m \leq f_t < 0$. Liquidity makers with valuation $v_b \geq \max\{f_m, 1/2\}$ propose $P = 0$, and the liquidity takers with $v_s \leq \min\{-f_t, 1/2\}$ accept the limit order. The cum fee buy and sell prices are

$$\begin{cases} p_b = f_m \\ p_s = -f_t \end{cases} \quad (8)$$

- b. Under the fee structure $(f_m + z, f_t - z)$, $z \in \mathbb{Z}$, the liquidity maker proposes $P = -z$, leaving cum fee prices the same like in part (a).

With a tick size of one, exchanges mandate both the cum fee buy price p_b and the cum fee sell price p_s . Exchanges can no longer mandate p_b and p_s after we reduce the tick size (see Section 8). Yet a tick size of one conveys

⁸ We thank Laura Cardella for helping us confirm this claim.

the model's economic intuition in the simplest way. In addition, Weyl (2010) points out that one of three main features of the two-sided platform models is to assume that platforms mandate the prices of both sides.

The multiplicity of equivalent fee breakdowns in part (b) of Lemma 2 follows Foucault, Kadan, and Kandel (2013). A one-tick increase in the make fee and a one-tick decrease in the take fee does not change the equilibrium outcome. What essentially matters for liquidity makers and takers, and thus for the exchange, is cum fee buy and sell prices. To avoid a tedious discussion of fee structures that achieve the same equilibrium outcome, we consider each exchange's decision variables as cum fee buy and sell prices in Sections 4–7.

4. Vertical Product Differentiation and Liquidity Makers' Segmentation

In this section, we consider the liquidity maker's and the liquidity taker's choices under exogenous fee structures. Subsection 4.1 shows the nonneutrality of the fee structure creates product differentiation for otherwise identical exchanges. Subsection 4.2 demonstrates liquidity makers' choice among exchanges.

4.1 Product differentiation

In this subsection, we show that the nonneutrality of the fee structure creates product differentiation for otherwise identical exchanges. The marginal liquidity taker who accepts the limit order on exchange i has valuation:

$$\hat{v}_s^i \equiv \min \left\{ p_s^i, \frac{1}{2} \right\}. \quad (9)$$

The liquidity taker accepts the buy limit order with probability:

$$q_s^i = \Pr(v_s \leq \hat{v}_s^i) = 2 \cdot \hat{v}_s^i = 2 \cdot \min \left\{ p_s^i, \frac{1}{2} \right\}. \quad (10)$$

A higher cum fee sell price implies a higher execution probability for the liquidity maker.

The liquidity maker's expected surplus for choosing exchange i is

$$BS^i = (v_b - p_b^i) \cdot q_s^i = 2 \cdot (v_b - p_b^i) \cdot \min \left\{ p_s^i, \frac{1}{2} \right\}. \quad (11)$$

All else being equal, Equation (11) shows that a liquidity maker prefers an exchange with a higher cum fee sell price because of the higher probability of realizing gains. The exchange with the higher cum fee sell price is of higher quality. Such vertical product differentiation is the fundamental driver for fee dispersion and market fragmentation.

Product differentiation across exchanges crucially depends on the nonneutrality of the fee structure. If end users can neutralize fees, exchanges are homogeneous and compete along one dimension, the total fee. Once end users cannot neutralize fees, price competition becomes two-dimensional, and an operator is able to set the price and quality of the product simultaneously.

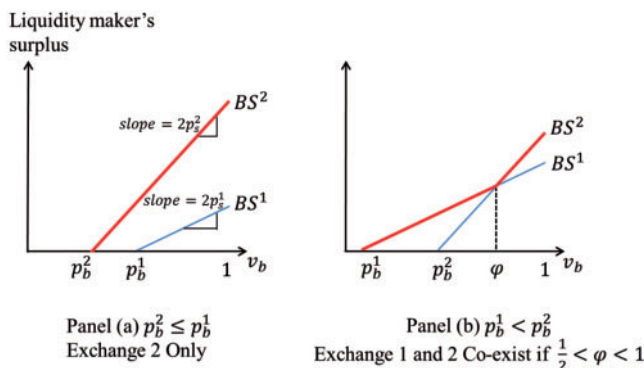


Figure 2

Liquidity maker's surplus under two exchanges

This figure shows the liquidity maker's surplus when choosing between two exchanges. The x -axis indicates the liquidity maker's valuation, and the y -axis indicates the liquidity maker's trading surplus. Without loss of generality, we assume that the cum fee sell price in exchange 1, p_s^1 , is lower than the cum fee sell price in Exchange 2, p_s^2 . BS^1 and BS^2 depict the liquidity maker's surplus when choosing exchange 1 and exchange 2, respectively. The thick lines depict the liquidity maker's choice, which is the upper envelope of the two surplus curves. φ is the cutoff value for the marginal liquidity maker.

4.2 Liquidity makers' segmentation across two exchanges

In this subsection, we consider the segmentation of the whole distribution of liquidity makers for two exchanges. A liquidity maker's expected surpluses for choosing exchange i is

$$BS^i = (v_b - p_b^i) \cdot q_s^i = 2 \cdot (v_b - p_b^i) \cdot p_s^i, i = 1, 2 \quad (12)$$

These equalities follow from $\hat{v}_s^i = p_s^i$ ($i = 1, 2$), because neither exchange would set $p_s^i > \frac{1}{2}$.

When $p_s^1 = p_s^2$, $BS^1 \geq BS^2$ if and only if $p_b^1 \leq p_b^2$. Without loss of generality, suppose that $p_s^1 < p_s^2$, which implies that exchange 1 is of low quality and exchange 2 is of high quality. Figure 2 shows the liquidity maker's surpluses in each exchange.

When $p_b^1 \geq p_b^2$, panel A of Figure 2 shows that $BS^1 \leq BS^2$ for any $v_b \geq p_b^2$, and no liquidity maker chooses exchange 1.⁹ In panel B of Figure 2, when $p_b^1 < p_b^2$, there is a unique intersection:

$$\varphi \equiv \frac{p_b^2 \cdot p_s^2 - p_b^1 \cdot p_s^1}{p_s^2 - p_s^1} \quad (13)$$

and $BS^1 \leq BS^2$ for any $v_b \geq \varphi$. Liquidity makers with a valuation higher than φ choose high-quality exchange 2 and liquidity makers with a valuation lower than

⁹ In Table 1, we show that exchanges can have the same fees on one side but unequal fees on the other side. One reason is that exchanges can also set their own criteria for traders to obtain a certain level of fees or rebates. Such criteria can lead to price discrimination within the same exchange; however, we focus on operators using multiple exchanges to conduct price discrimination.

φ choose low-quality exchange 1. Exchanges 1 and 2 coexist when $\frac{1}{2} < \varphi < 1$. All things being equal, liquidity makers prefer the high-quality exchange 2. Yet liquidity makers with larger gains from execution care more about execution probability than those with smaller gains. This trader heterogeneity allows an exchange with higher cum fee buy and sell prices to coexist with an exchange with lower cum fee buy and sell prices.

5. Vertical Differentiation with a Monopolist Exchange

In this section, we characterize a monopoly operator's choice of the number of exchanges to offer, as well as their fee structures. Proposition 2 shows the second-degree price discrimination.

Proposition 2 (Number of exchanges established by a monopoly operator and associated fee structures). Under tick size constraints in Equation (1), for a monopoly operator who operates K exchanges, the optimal cum fee buy and sell prices in each exchange i are

$$p_b^i = \frac{1}{2} + \frac{i}{2(2K+1)} p_s^i = \frac{i}{2K+1} \text{ with } 1 \leq i \leq K. \quad (14)$$

A liquidity maker with valuation $v_b \in [\varphi_{i-1}, \varphi_i]$ chooses exchange i , where

$$\varphi^i = \begin{cases} p_b^1 & \text{for } i=0 \\ \frac{p_b^{i+1} p_s^{i+1} - p_b^i p_s^i}{p_s^{i+1} - p_s^i} & \text{for } 1 \leq i \leq K-1 \\ 1 & \text{for } i=K. \end{cases}$$

The liquidity maker's expected surplus, the liquidity taker's expected surplus, and the monopoly operator's expected profit are

$$BS^M(K) = \frac{K \cdot (K+1)}{3(2K+1)^2}, SS^M(K) = \frac{K \cdot (K+1)}{3(2K+1)^2}, \Pi^M(K) = \frac{2K \cdot (K+1)}{3(2K+1)^2}, \quad (15)$$

respectively, which all increase in K .

If opening a new exchange requires a fixed cost c , the number of exchanges opened by the monopoly operator is

$$\bar{K} = \max \left\{ K \in \mathbb{N} \mid \frac{4K}{3(2K+1)^2(2K-1)^2} \geq c \right\}.$$

The monopoly operator's strategy follows standard menu pricing under second-degree price discrimination (Mussa and Rosen 1978). The monopoly operator uses K exchanges to segment the liquidity makers into $K+1$ groups. The group with the lowest gains from execution does not participate. Among the rest of the K groups, liquidity makers with higher gains from execution tend to choose exchanges with higher cum fee buy and sell prices.

Our model offers two unique features compared with the standard menu pricing model. First, the quality of the exchange is the cum fee sell price, and the exchanges use the terms of the trade offered to liquidity takers to screen liquidity makers. This provides explanatory power for the puzzle raised by (Foucault, 2012, p. 20) that “it is not clear however how the differentiation of make/take fees suffices to screen different types of investors.” Second, the source of the price discrimination is nonneutrality led by the discrete price, because price discrimination does not exist when liquidity makers and takers can neutralize the fee breakdowns.

Although the monopoly operator’s profit increases with the number of exchanges, the operator’s marginal profit decreases. Any fixed cost incurred by the monopoly operator for establishing an exchange thus constrains the number of exchanges.

Interestingly, Equation (15) indicates that both the liquidity maker’s and taker’s expected surpluses increase with the number of exchanges. The welfare gain originates from a higher participation rate: more exchanges create more cum fee price levels within the tick. As K goes to infinity, the lowest cum fee buy price across all exchanges, $\frac{K+1}{2K+1}$, approaches $\frac{1}{2}$, and almost all liquidity makers will participate. The creation of new cum fee price levels reduces the inefficiency created by discrete tick size, and increases the expected trading gains for all parties.

Table 2 provides an example of such second-degree price discrimination. Column (1) shows that a monopoly operator with one exchange sets $p_b^M = \frac{2}{3}$ and $p_s^M = \frac{1}{3}$. Liquidity makers with valuation in $[\frac{2}{3}, 1]$ and liquidity takers with valuation in $[0, \frac{1}{3}]$ participate. The operator has a profit of $\frac{4}{27}$. Columns 2 and 3 show that the operator can screen liquidity makers by setting up two exchanges. The low-quality exchange, with $p_b^1 = \frac{3}{5}$ and $p_s^1 = \frac{1}{5}$, offers an execution probability of $\frac{2}{5}$ and attracts liquidity makers with lower gains from execution ($v_b \in [\frac{3}{5}, \frac{4}{5})$). The high-quality exchange, with $p_b^2 = \frac{7}{10}$ and $p_s^2 = \frac{2}{5}$, offers an execution probability of $\frac{4}{5}$ and attracts liquidity makers with higher gains from execution ($v_b \in [\frac{4}{5}, 1]$). This second-degree price discrimination increases the operator’s profit from $\frac{4}{27}$ to $\frac{4}{25}$. The expected surplus for both liquidity makers and takers increases from $\frac{2}{27}$ to $\frac{2}{25}$. In addition, the expected trading volume increases from $\frac{12}{27}$ to $\frac{12}{25}$. This provides justification for the expected welfare gains for all parties.

6. Vertical Differentiation with Two Competing Exchanges

In this section, we consider the case of two competing operators, each of whom establishes one exchange. In Subsection 6.1, we establish the nonexistence of the pure-strategy equilibrium. In Subsection 6.2, we show the existence of a symmetric mixed-strategy equilibrium and that any mixed-strategy equilibrium leads to positive profit.

Table 2
Second-degree price discrimination by a monopoly operator

	One exchange	Two exchanges	
		Low quality	High quality
	(1)	(2)	(3)
<i>A. Cum fee prices</i>			
<i>Cum fee sell price</i>	1/3	1/5	2/5
<i>Execution probability (quality)</i>	2/3	2/5	4/5
<i>Cum fee buy price</i>	2/3	3/5	7/10
<i>B. Range of participants</i>			
<i>Liquidity maker participation</i>	[2/3, 1]	[3/5, 4/5]	[4/5, 1]
<i>Liquidity taker participation</i>	[0, 1/3]	[0, 1/5]	[1/5, 2/5]
<i>C. Welfare</i>			
<i>Operator</i>	4/27	4/25	
<i>Liquidity maker</i>	2/27	2/25	
<i>Liquidity taker</i>	2/27	2/25	

This table shows the market outcomes when a monopoly establishes one exchange and two exchanges. Panel A displays the cum fee sell and buy prices in each exchange. Panel B displays the valuation ranges of liquidity makers and liquidity takers who participate in the exchange, and their corresponding choices of the exchanges. Panel C shows the expected welfare for the operator, liquidity makers, and liquidity takers.

6.1 No pure-strategy equilibrium

In Proposition 3, we show that tick size constraints destroy not only the Bertrand equilibrium but also *any* pure-strategy equilibrium.

Proposition 3 (No pure-strategy equilibrium). Under tick size constraints in Equation (1), there exists no pure-strategy equilibrium when there is competition between two exchanges.

Similar to the Bertrand competition, no pure-strategy equilibrium exists if exchanges have unequal profits, because the lower-profit exchange can increase its profit by undercutting the higher-profit exchange’s cum fee buy price by ε and mimicking its cum fee sell price. Also, no pure-strategy equilibrium exists if operators have positive and equal profits. If identical profits are led by identical price structures, one exchange can increase its expected profit by undercutting its rival’s cum fee buy price by ε and mimicking its rival’s cum fee sell price.

When price competition is two-dimensional, it is possible for the two exchanges to have different price structures but the same total profits, and we demonstrate that no pure-strategy equilibrium exists in such a case.¹⁰ Because each operator can establish only one exchange, we refer to the fee structure of the exchange established by operator i as (p_b^i, p_s^i) . Without loss of generality, suppose that, initially, exchange 2 has higher quality than exchange 1 ($p_s^2 > p_s^1 = \gamma$). Figure 2 shows that exchange 2 must have a higher cum fee buy price than exchange 1 ($p_b^2 > p_b^1 = \delta$), and that liquidity makers with high valuations choose exchange 2 while liquidity makers with low valuation choose exchange 1. Exchange 2 then can increase its profits by reducing the cum fee sell

¹⁰ Numerical examples of such cases are available from the authors on request.

price to γ and the cum fee buy price to $\delta - \varepsilon$. This deviation steals all exchange 1's business using the ε price concession. In addition, exchange 2 retains all liquidity makers who originally chose exchange 2, because the new fee structure in exchange 2 offers the same quality to liquidity makers as exchange 1 but charges a lower cum fee buy price. When ε is sufficiently small, the loss of original customers from exchange 1 can be easily compensated by preserving exchange 2's liquidity makers. Therefore, no pure-strategy equilibrium exists when two exchanges have different pricing structures but the same total profits.

Surprisingly, no pure-strategy equilibrium exists even if both operators make zero profits. Two possible scenarios lead to the zero-profit outcome: (1) no trading or (2) the cum fee buy and sell prices are equal. Case (1) is trivial, because one of the two operators can deviate profitably by facilitating some trades. Case (2) involves two types of deviation.

If $p_b^1 = p_s^1 \geq \frac{1}{2}$, exchange 2 can deviate profitably by setting $p_b^2 = p_b^1 - \mu\varepsilon$, and $p_s^2 = p_s^1 - \varepsilon$ with $\varepsilon > 0$ and $0 < \mu < 1$. When $p_b^1 = p_s^1 > \frac{1}{2}$, exchange 2 reduces the cum fee sell price but not the execution probability, because any liquidity taker accepts $p_s^2 = \frac{1}{2}$. A lower cum fee buy price attracts liquidity makers with a valuation above p_b^2 to exchange 2. When $p_b^1 = p_s^1 = \frac{1}{2}$, such a deviation attracts liquidity makers with lower gains from execution by reducing the execution probability and the cum fee buy price, which is illustrated in panel A of Figure 3. If $p_b^1 = p_s^1 < \frac{1}{2}$, panel B of Figure 3 shows that exchange 2 can deviate profitably by setting $p_b^2 = p_b^1 + \varepsilon$ and $p_s^2 = p_s^1 + \mu \cdot \varepsilon$. Such a deviation increases the execution probability and the cum fee buy price, which attracts liquidity makers expecting higher gains from execution.

In our model, the simultaneous choice of price and quality destroys the pure-strategy equilibrium. The model would generate a standard pure-strategy equilibrium in traditional price-quality games (Shaked and Sutton 1982; Tirole 1988) if operators choose the take fee in the first stage and the make fee in the second stage.¹¹ In manufacturing industries, it is natural to assume that the product quality decision is made prior to its price determination because firms can adjust prices faster than quality (Tirole 1988). Yet the quality in our model, determined purely by the cum fee sell price, can be adjusted as easily as the cum fee buy price.

6.2 Mixed-strategy equilibrium

The nonexistence of the pure-strategy equilibrium motivates us to investigate mixed-strategy equilibria. Proposition 4 demonstrates the existence of a symmetric mixed-strategy equilibrium. Varian (1980) suggests that the mixed-strategy equilibrium justifies the spatial price dispersion and temporal price dispersion. From this perspective, we provide the first theoretical interpretation for diverse fee structures across exchanges and their frequent changes. As

¹¹ For brevity, we do not report these results.

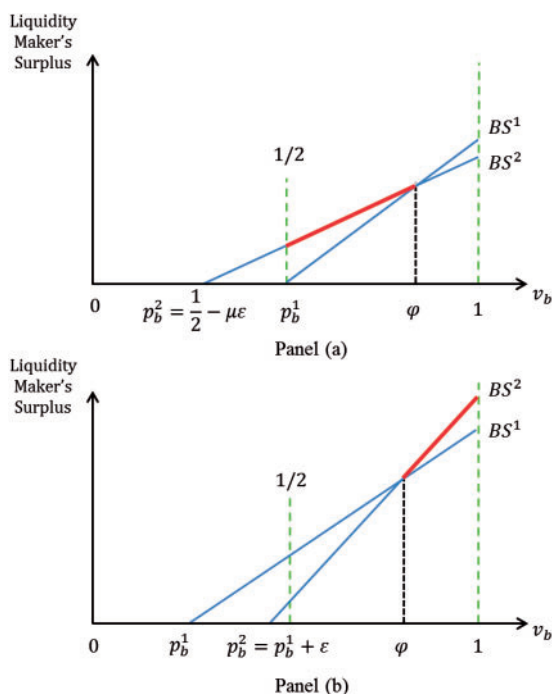


Figure 3

Two types of deviations from the Bertrand equilibrium

This figure shows two types of profitable deviation for exchange 2 when exchange 1 and Exchange 2 start with zero profits. The x -axis indicates the liquidity maker's valuation, and the y -axis indicates the liquidity maker's trading surplus. BS^1 and BS^2 depict the liquidity maker's surplus when choosing exchange 1 and exchange 2, respectively. In panel A, exchange 1 sets $p_b^1 = p_s^1 = \frac{1}{2}$. Exchange 2 can profitably deviate by decreasing the cum fee buy price to $p_b^2 = \frac{1}{2} - \mu\epsilon$ and the cum fee sell price to $p_s^2 = \frac{1}{2} - \epsilon$. In panel B, exchange 1 sets $p_b^1 = p_s^1 < \frac{1}{2}$. Exchange 2 can profitably deviate by increasing the cum fee buy price to $p_b^2 = p_b^1 + \epsilon$ and the cum fee sell price to $p_s^2 = p_s^1 + \mu\epsilon$. $\epsilon > 0$ and $0 < \mu < 1$ in both panels. The liquidity maker chooses the exchange that offers the highest surplus. The thick lines depict the liquidity makers who choose exchange 2. φ is the cutoff value for the marginal liquidity maker.

indicated by an SEC statement, “these exchanges compete vigorously on price which leads to some rather complicated fee schedule that can change from month to month, making it a near full-time job to keep track of them all.”¹²

Proposition 4 (Mixed-strategy equilibrium). Under the tick size constraints in Equation (1):

- a A symmetric mixed-strategy equilibrium exists; and
- b In equilibrium, both exchanges earn strictly positive profits.

¹² See Richard Holley, SEC Trading and Division, Statement in Panel Discussion at SEC Equality Market Structure Advisory Committee Meeting on October 27, 2015, available at <https://www.sec.gov/news/otherwebcasts/2015/equality-market-structure-advisory-committee-102715.shtml>.

The driver of our mixed-strategy equilibrium significantly differs from those in the canonical one-dimensional mixed-strategy equilibrium, which generally involves frictions that prevent customers from finding the best price (Rosenthal 1980; Varian 1980; Burdett and Judd 1983), or cost for firms to advertise their prices (Butters 1977; Baye and Morgan 2001). Without these frictions, firms with the lowest price would attract all customers. In our model, the liquidity maker and taker fully optimize their choices, know the fees, and exchanges do not incur any costs to advertise their make/take fees to traders. The driving force behind the mixed-strategy equilibrium is two-dimensional price competition.

When price competition is one-dimensional, firms with the lowest price attract all customers who are aware of the lowest price. The winner-take-all property simplifies the solution for the mixed-strategy equilibrium. When price competition is two-dimensional, liquidity makers have no consensus on the best prices. The profit function of an exchange becomes a complex double integral over makers' and takers' valuations conditional on the other exchange's strategy. It is a daunting task to find analytical solutions for a mixed-strategy equilibrium with two-dimensional price competition. Nevertheless, the proof of Proposition 4 shows the existence of a symmetric mixed-strategy equilibrium.

The positive profit in the symmetric mixed-strategy equilibrium provides the rationale for the entries of new exchanges. This result arises from the two-sided feature of the markets caused by the tick size regulation. When tick size is zero, as shown in Section 3, the markets are one-sided. Hence, competition between two exchanges drives their profits to zero (Colliard and Foucault 2012), which implies that any positive cost involved in establishing a new stock exchange would deter entry. In reality, however, we continue to witness "the formation of new exchanges to experiment different price structures."¹³ When the tick size is positive, the markets become two-sided, and competition between two exchanges does not drive profits to zero.

We cannot express the number of competing exchanges as a function of the fixed cost c , because the amount of profits do not have analytical solutions. However, we are able to characterize the free entry equilibria with c equal to 0 in next section.

7. Free Entry Equilibria

In this section, we consider free entry equilibria. The ability to create a new exchange with zero costs restores the pure-strategy equilibrium. The search for the pure-strategy equilibrium starts with Lemma 3.

¹³ See Richard Holley, SEC Trading and Division, Statement in Panel Discussion at SEC Equality Market Structure Advisory Committee Meeting on October 27, 2015, available at <https://www.sec.gov/news/otherwebcasts/2015/equity-market-structure-advisory-committee-102715.shtml>.

Lemma 3 (Ideal breakeven cum fee buy and sell prices for liquidity makers). For a liquidity maker with valuation v_b , a fee structure that offers cum fee buy and sell prices, $p_b = p_s = \frac{v_b}{2}$, maximizes the expected trading surplus, under the exchange's breakeven constraints $p_b \geq p_s$.

Lemma 3 shows that, under the breakeven constraints for exchanges, a liquidity maker with v_b strictly prefers an exchange with $p_b = p_s = \frac{v_b}{2}$. As liquidity makers' valuation v_b is uniformly distributed on $[\frac{1}{2}, 1]$, the ideal breakeven cum fee buy and sell prices for all liquidity makers requires a dense set of cum fee buy and sell prices in the interval $[\frac{1}{4}, \frac{1}{2}]$. Lemma 4 shows that the failure to generate such a dense set leads to an opportunity for positive profits.

Lemma 4 (Opportunity for positive profits when cum fee sell prices are not dense in $[\frac{1}{4}, \frac{1}{2}]$). Suppose that all exchanges at least break even. Denote the set of all cum fee sell prices of all exchanges as P_s . If $\exists a, b$, where $\frac{1}{2} < 2a < 2b < 1$ such that $P_s \cap (a, b) = \emptyset$, then an exchange can earn a positive profit by setting $p_s = \frac{a+b}{2}$ and a $p_b \in (\frac{a+b}{2}, \frac{a^2+b^2}{a+b})$, which has $T = p_b - p_s > 0$ and attracts some liquidity makers with $v_b \in (2a, 2b)$.

In Proposition 3, we show that no pure-strategy equilibrium exists even when both operators have zero profit. Lemma 4 provides an alternative way to understand this result. Exchange operators have profit opportunities as long as a positive measure of liquidity makers does not attain the liquidity makers' optimal cum fee prices. When the number of exchanges is finite, there must be a gap in cum fee sell prices. The gap then allows an operator to attract some liquidity makers by moving closer to their optimal cum fee prices, even if the operator charges a positive total fee. In Section 6, no pure-strategy equilibrium exists because the finite number of fee structures can never create a dense set of prices. Free entry restores the pure-strategy equilibrium because operators can create a dense set of prices in $[\frac{1}{4}, \frac{1}{2}]$ by establishing new exchanges.

Proposition 5 (Necessary and sufficient condition for free entry equilibria). Under the tick size constraints in Equation (1), the set of fee structures is an equilibrium if and only if $\forall x \in [\frac{1}{4}, \frac{1}{2}]$, $\forall \underline{i}$, and $\varepsilon > 0$, $\exists$ Operator $\bar{i} \neq \underline{i}$ and its exchange s.t. $x + \varepsilon > p_b^{\bar{i},k} = p_s^{\bar{i},k} > x - \varepsilon$.

Proposition 5 reveals two necessary conditions of the free entry equilibria. First, all exchanges must earn zero profit. Otherwise, a new entrant can simply establish the same fee structures as the profitable exchange and share part of its profit. It follows that $p_b^{i,k} = p_s^{i,k}$ for any active exchange (i, k) . Second, any interval in $[\frac{1}{4}, \frac{1}{2}]$ must contain at least two exchanges belonging to two different operators. If there is no exchange in the interval, an operator can obtain positive profit by creating a new exchange with its fee structure, as outlined in Lemma 4. If all exchanges in the interval belong to the same operator, the operator can obtain positive profit by adopting the fee structure outlined in Lemma 4 in one exchange and closing the other exchanges in the interval.

These two conditions are also sufficient for pure-strategy equilibria. No exchange operator can increase its profit by adjusting the fee structures or adding new exchanges, because the exchange fee structures from competing operators have already served liquidity makers with their ideal quality with the lowest possible cum fee buy price.

Proposition 5, as a reference point that cannot be obtained in reality, reveals that fee competition has two dimensions in free entry equilibria. The first dimension reduces the total fee toward zero. The second dimension, which is unique in our model, is to restore continuous prices for traders. In this sense, we can consider make and take fees as a financial innovation to span continuous price levels, when regulators mandate discrete trading prices. In reality, establishing an exchange requires fixed costs, and the number of exchanges is finite. The inability to create continuous price then destroys pure-strategy free entry equilibrium and provides an alternative way to understand the frequent change of fees.

8. Robustness Checks: Smaller Tick Size

As robustness checks, we reduce the tick size to $\frac{1}{N}$, where N is a natural number greater than one. As a liquidity maker has more price grids to choose from, exchange operators' make/take fees no longer mandate unique cum fee buy and sell prices. Instead, the fee structure of exchange k owned by operator i , $(f_m^{i,k}, f_t^{i,k})$ offers a set of cum fee buy and sell prices $\left\{ \left(p_b^{i,k,n}, p_s^{i,k,n} \right) \middle| p_b^{i,k,n} = \frac{n}{N} + f_m^{i,k}, p_s^{i,k,n} = \frac{n}{N} - f_t^{i,k}, n \in \mathbb{Z} \right\}$. A liquidity maker chooses (i, k, n) to maximize the expected surplus $(v_b - p_b^{i,k,n}) \times \Pr(v_s \leq p_s^{i,k,n})$, and the exchange operators choose the number of exchanges and fee structures to maximize profit. The competition over a pricing grid $(p_b^{i,k,n}, p_s^{i,k,n})$ makes the solution with $N > 1$ very complex and often intractable. Predictions under this relaxed setting, however, are qualitatively similar to those under $N=1$. In Subsection 8.1, we show the second-degree price discrimination. In Subsection 8.2, we show the nonexistence of a pure-strategy equilibrium with a finite number of operators, each with finite exchanges. In Subsection 8.3, we analyze the free entry equilibria. We present all proofs for this section in the Online Appendix.

8.1 Second-degree price discrimination with a monopoly operator

Our analysis of second-degree price discrimination starts with a monopoly operator with one exchange. First, we consider a liquidity maker's choice after observing the fee structure of the exchange, (f_m, f_t) . Proposition 6 shows whether or not a liquidity maker decides to participate in trading and, if participating, the optimal limit order price proposed.

Proposition 6 (The limit order price proposed by the liquidity maker under one exchange). When tick size is $\frac{1}{N}$:

- The liquidity maker submits no limit order when $v_b < f_m + \frac{[N \bullet f_t]}{N}$, while a liquidity maker submits a limit order at price $P = \frac{n(v_b, f_m, f_t)}{N}$ otherwise. Here, $n(v_b, f_m, f_t) = \left\lfloor N \left(\frac{v_b + T}{2} - f_m \right) \right\rfloor$, where $\lceil x \rceil$ denotes the smallest integer greater than or equal to x and $\lfloor x \rfloor$ denotes the nearest integer to x .
- Compared with fee structure (f_m, f_t) , fee structure $(f_m + \frac{z}{N}, f_t - \frac{z}{N})$, $z \in \mathbb{Z}$ shifts the limit order price proposed by the liquidity maker by $-z$ ticks, but leads to the same cum fee buy/sell prices as those under fee structure (f_m, f_t) .

Part (a) of Proposition 6 shows that a liquidity maker proposes a price that is nearest to the unconstrained optimal nominal price in the pricing grid, $\frac{v_b + T}{2} - f_m$. The fee breakdown is generally not neutral when tick size is $\frac{1}{N}$. For example, holding T fixed, a small increase in f_m may not change the nominal price $\frac{\lfloor N \left(\frac{v_b + T}{2} - f_m \right) \rfloor}{N}$, which leads to an increase in the cum fee buy price $\frac{\lfloor N \left(\frac{v_b + T}{2} - f_m \right) \rfloor}{N} + f_m$. Part (b) of Proposition 6 shows that fee breakdowns in the multiple of ticks is neutral. For a given total fee, if the exchange increases its make fee by, say, one tick, then a liquidity maker can neutralize the effect of this increase by proposing a limit order price that is one tick lower, leaving the cum fee buy and sell prices unchanged. Taking together, Proposition 6 reconfirms that discrete price creates nonneutrality: when the tick size is zero, a liquidity maker is able to neutralize any fee breakdown.

Proposition 6 demonstrates that a liquidity maker with higher gains from execution tends to propose a higher limit order price, but the limit order price does not strictly increase in v_b . Discrete tick size may force liquidity makers who would propose a heterogeneous nominal price under continuous price to propose the same nominal price. The second-degree price discrimination under multiple ticks involves screening liquidity makers who have heterogeneous valuations but quote the same price.

The liquidity maker's optimal limit order price involves the nearest integer function of the fees, and participation constraints involve the ceiling function. These two noncontinuous functions make it extremely difficult to characterize the optimal fee structure for the operator. The solution for optimal fees with one exchange requires complex mathematical construction to circumvent the noncontinuity, and we present the solution in the Online Appendix. When the operator establishes more than one exchange, a liquidity maker's choice involves multiple nearest integer functions of the fees, making it almost impossible to express the fee structures in closed form. Therefore, we conduct a numerical search for the highest profit for a monopoly operator with

two exchanges and present the results for $N=2$, 4, and 8 as examples in Table 3.¹⁴

Table 3 shows that the $N=2$ case is identical to the $N=1$ case.¹⁵ The $N=4$ and $N=8$ cases illustrate two interesting features. First, two vertically differentiated exchanges increase the monopoly operator's profits. For example, for $N=4$, the low-quality exchange sets the take fee at 0.0340 and the high-quality exchange sets the take fee at -0.1595 .¹⁶ The second-degree price discrimination, which involves charging a low make fee (0.4095) for the low-quality exchange and a high make fee (0.5840) for the high-quality exchange, increases the monopoly operator's total profit from 0.1406 to 0.1445 (0.1107+0.0338).

Second, at the same nominal price, liquidity makers who choose the high-quality exchange have higher valuation. Yet liquidity makers rotate between the low-quality and the high-quality exchanges as their valuation increases. For example, under tick size $\frac{1}{4}$, a liquidity maker with valuation $v_b \in [0.5000, 0.6313]$ proposes a limit order at $P=0$ on the low-quality exchange. A liquidity maker with valuation $v_b \in [0.6313, 0.7562]$ proposes the same price in the high-quality exchange. A further increase in valuation to $v_b \in [0.7562, 1.000]$ leads a liquidity maker to switch back to the low-quality exchange with an increased limit order price of $P = \frac{1}{4}$.

The rotation of the exchange choice, however, does not imply horizontal differentiation between exchanges. Exchanges in our model are only vertically differentiated: all else being equal, all liquidity makers prefer an exchange with a lower take fee. Exchange rotation is driven by the definitions of the high and low valuation types. Unlike standard models of vertical differentiation, the high and low types in our game are defined on each price grid. Table 3 shows that, at the same limit order price, liquidity makers who choose the high-quality exchange have higher valuation than those who choose the low-quality exchange. The nature of such second-degree price discrimination is to screen liquidity makers with relatively high and low valuations on the same nominal price. Therefore, the price discrimination with multiple ticks is an extension of our model with only one tick, in which all liquidity makers quote the same price.

¹⁴ We restrict the level of the take fee to $[-\frac{1}{N}, 0]$. We set the fee increment of our search to 0.0005 and examine combinations of fee structures $(f_m^1, f_t^1, f_m^2, f_t^2)$ within the feasible range. Each combination leads to two series of implied pricing grids for liquidity makers. We then randomly draw a liquidity maker from a uniform distribution from $[\frac{1}{2}, 1]$ who chooses from any price generated by the fee structures of the two exchanges. If a liquidity maker proposes a nominal price in an exchange, we can calculate the cum fee sell price and the participation probability of the liquidity taker. Averaging over all liquidity makers, we obtain the expected profit for each fee structure. We present the fee structure (combinations) with the highest profit in Table 3.

¹⁵ It is because tick sizes are still large enough for make and take fees to mandate cum fee buy and sell prices.

¹⁶ Because a liquidity maker always chooses the exchange with the lower take fee if both exchanges set the same make fee, the exchange with the lower take fee is of higher quality.

Table 3
Exchange fee structures and liquidity makers' segmentation under various tick sizes

	(1)	$N=2$ (2)	(3)	(4)	$N=4$ (5)	(6)	(7)	$N=8$ (8)	(9)
	One exchange	Two exchanges		One exchange	Two exchanges		One exchange	Two exchanges	
		Exchange 1	Exchange 2		Exchange 1	Exchange 2		Exchange 1	Exchange 2
<i>A. Fee structure of exchanges</i>									
<i>Make fee</i>	0.6667	0.6000	0.7000	0.3750	0.4095	0.5840	0.3750	0.4165	0.4705
<i>Take fee</i>	−0.3333	−0.2000	−0.4000	0.0000	−0.0340	−0.1595	0.0000	−0.0339	−0.0961
<i>Total fee</i>	0.3333	0.4000	0.3000	0.3750	0.3755	0.4245	0.3750	0.3826	0.3744
<i>Profit</i>	0.1481	0.0640	0.0960	0.1406	0.1107	0.0338	0.1406	0.0421	0.0994
<i>B. Liquidity makers' segmentation</i>									
$P=0$	0.6667–1.0000	0.6000–0.8000	0.8000–1.0000		0.5000–0.6313	0.6313–0.7562			0.5000–0.6501
$P=1/8$							0.5000–0.7500	0.6501–0.7334	0.7334–0.9164
$P=2/8$				0.6250–1.0000	0.7562–1.0000		0.7500–1.0000	0.9164–1.0000	
$P=3/8$									
$P=4/8$									

This table provides an example of fee structures set by the monopoly exchange operator, as well as and the segmentation of liquidity makers under various tick sizes. Columns 1–3, 4–6, and 7–9 display the market outcomes under tick sizes of $\frac{1}{2}$, $\frac{1}{4}$, and $\frac{1}{8}$, respectively. Panel A lists the optimal fee structures. The rows in panel B indicate the limit order price that the liquidity maker proposes and the column indicates the liquidity maker's choice of exchange conditional on the valuation.

8.2 No pure-strategy equilibrium with more than one operator

In this subsection, we extend Proposition 3 in three dimensions. We consider general tick size $\frac{1}{N}$, allow two or more operators, and allow each operator to establish more than one exchange. Still, no pure-strategy equilibrium exists.

Proposition 7 (No pure-strategy equilibrium with competing operators).

Under tick size $\frac{1}{N}$ ($N \geq 1$), there exists no pure-strategy equilibrium when there are $I \geq 2$ operators with $K \geq 1$ exchanges each.

The proof for Proposition 7 is more complex in three dimensions: multiple ticks, multiple operators, and multiple exchanges, but it follows the same intuition as the proof of Proposition 3 and Lemma 4. Positive profit leads to profitable deviation by undercutting one side of the fees and keeping the other side uncharged. Zero profits lead to profitable deviation to create cum fee sell prices that are not currently available, because a finite number of exchanges cannot span the ideal breakeven cum fee prices for a continuum of liquidity makers.

8.3 Free entry equilibria

In this subsection, we analyze the comparative statics of the free entry equilibria with respect to the tick size $\frac{1}{N}$. The interaction between fee structures and the size of the tick can be used to evaluate two SEC pilot programs. On October 3, 2016, the SEC implemented a pilot program to increase the tick size from 1 cent to 5 cents for 1,200 small stocks. The SEC also began to discuss a make/take fee pilot program to reduce the cap for take fees (SEC 2016). Our model offers insights that help to coordinate these two pilot programs.

Corollary 1 (Free entry equilibria with multiple ticks). Under the tick size constraints in Equation (1'), the set of fee structures is an equilibrium if and only if $\forall x \in [\frac{1}{4}, \frac{1}{2}]$, $\forall \underline{i}$, and $\varepsilon > 0, \exists$. Operator $\bar{i} \neq \underline{i}$ s.t. $x + \varepsilon > p_b^{\bar{i},k,n} = p_s^{\bar{i},k,n} > x - \varepsilon$.

Corollary 1 follows the same intuition as Proposition 5, except that exchanges use a pricing grid to span the price interval $[\frac{1}{4}, \frac{1}{2}]$. The take fee cap does not impose a binding constraint, because liquidity makers can neutralize fee breakdowns in multiple ticks. In reality, the take fee cap matters because the Rule 611 protects the best nominal price but not the cum fee price. Among the fee structures that lead to the same cum fee prices, the one with the maximum take fee offers the best nominal price for liquidity makers. Rule 610, therefore, imposes a take fee cap of 0.3 cents to prevent exchanges from achieving artificially better nominal prices.

In Corollary 2, we refine the equilibria to fee structures that offer the best nominal price.

Corollary 2 (Free entry equilibrium with the best nominal price). Denote $\bar{f}_i$ as the cap of take fee. When $N \geq 4$,

a. The equilibrium fee structures for active exchanges are

$$\left\{ (f_m^{i,k}, f_t^{i,k}) \mid f_m^{i,k} = -f_t^{i,k} \in \left[-\bar{f}_t, -\bar{f}_t + \frac{1}{N} \right] \right\}.$$

- b. An increase in tick size increases the dispersion of the fee structure, $\max(f_m^{i,k}) - \min(f_m^{i,k})$ and the maximum make fee $\max(f_m^{i,k})$.
- c. A decrease in the take fee cap increases the maximum make fee, but does not change the dispersion of fee structures.

Part (a) of Corollary 2 states that exchanges charge the take fee towards the cap in order to achieve more favorable nominal prices. When $N \geq 4$, the maximum take fee is exactly at the cap, and the minimum take fee is one tick below the maximum.¹⁷ When the take fee cap is positive, exchanges lean towards giving rebates to liquidity makers. The protection of nominal quotes, therefore, provides one explanation on why some exchanges charges the maximum take fee of 0.3 cents and offer aggressive rebates to liquidity makers.

Yet part (a) of Corollary 2 also explains why some other exchanges charge liquidity makers. In our model, the nature of the fee game is to span continuous price. This new mechanism differs from two alternative interpretations of fee structures: (1) operators use rebates to incentivize liquidity provision (Cardella, Hao, and Kalcheva 2015; Malinova and Park 2015) and (2) brokers route limit orders to exchanges with the highest rebates (Angel, Harris, and Spatt 2010, 2013; Battalio, Corwin, and Jennings 2016). Both explanations provide support for the use of rebates to liquidity makers, but they cannot explain why some operators charge liquidity makers.

In part (b) of Corollary 2, the fee dispersion is defined as the difference between the maximal make fee and the minimum make fee, because total fees are 0 in free entry equilibria. Part (b) shows that an increase in tick size increases fee dispersion. Because of the take fee cap, an increase in fee dispersion leads to an increase in the maximum make fee. Part (c) of Corollary 2 shows that a reduction in the take fee cap has no effect on fee dispersion, but the maximum charge to liquidity makers across all exchanges would increase.

Certainly, it is unrealistic to have infinitely many exchanges. We use Corollary 2 mainly as a reference point to provide further economic intuition on fee competition. In addition, Corollary 2 provides insights on the design of tick size and make/take fee pilot programs. As both an increase in tick size and a decrease in take fee lead to more aggressive charges to liquidity makers, it is important for policy makers to create a staggered sample for both pilot programs

¹⁷ Because we assume that liquidity makers' valuation is $[\frac{1}{2}, 1]$, fee dispersion is $\frac{1}{4}$ when $N < 4$. Yet the predicted fee dispersion is not unrealistic. Table 1 shows that the fee dispersion in reality is about 0.4 of a tick, which is in between the case in which the tick size is equal to 1, fee dispersion is $\frac{1}{4}$ of a tick, and the tick size is equal to $\frac{1}{2}$, when fee dispersion is $\frac{1}{2}$ of a tick.

in order to disentangle their relative impacts. The ideal sample contains 2-by-2 groups of stocks with a similar number of observations to (1) change tick size and take fee cap; (2) change only tick size; (3) change only the take fee cap; and (4) no change.

9. Conclusion

We show that discrete prices fragment U.S. stock exchanges and lead to dispersions in the fees they charge traders. When traders can quote a continuous price, the breakdown of the make/take fees is neutral and orders stream to the exchange with the lowest total fee. Under tick size constraints, fee breakdowns are no longer neutral, and exchanges compete on both make and take fees. The nonneutrality of fees allows a monopoly operator to establish multiple exchanges with heterogeneous fee structures for second-degree price discrimination. When multiple operators compete, tick size constraints destroy the pure-strategy equilibrium as long as the number of exchanges is finite, which rationalizes fee diversity and frequent fluctuations. In free entry equilibria, the competing exchanges must restore continuous cum fee price. Therefore, competition on fees and fragmentation among exchanges is a type of financial innovation in response to mandated discrete tick size.

Our model provides a new perspective to evaluate the proposal to ban the maker/taker pricing model. Since there is limited economic rationale for the complexity of fees, Lewis (2014) conjectures that such complexity may serve fraudulent practices (SEC 2015b). However, the nonexistence of pure strategy equilibrium provides an alternative interpretation of this complexity. Battalio, Corwin, and Jennings (2016) find that broker/dealers have a strong incentive to route customers' limit orders to the market offering the highest rebate, because brokers/dealers can keep such rebates. This conflict of interest leads to two policy proposals: (1) passing the rebate back to customers; and (2) eliminating the fees (Angel, Harris, and Spatt 2010, 2013). This paper reveals economic forces that favor the first solution.

Our findings challenge the proposed initiative to increase the tick size for small stocks to five cents. Advocates argue that it may increase market-making profits, support sell-side equity research, and, eventually, increase the number of IPOs (Weild, Kim, and Newport 2012). We show, however, that a wider tick size would increase the fee dispersion to create more continuous cum fee prices, and lead to a more fragmented market. Chao, Yao, and Ye (2017) provide empirical evidence that a larger tick size leads to more fragmented stock trading.

We show that nonneutral fee structures leads to market fragmentation and price dispersion. This insight can apply to other market frictions that destroy neutrality. For example, nonsurcharge provisions in credit card contracts discourage merchants from charging differentiated prices for payments using different credit cards, thereby preventing merchants and card-holders from neutralizing fee breakdowns. If merchants could impose surcharges on

transactions, credit card issuers would compete over the total fees. Our model implies that the removal of the nonsurcharge provision would reduce the variety of credit cards. Finally, we show that the simultaneous choice of price and quality destroys pure-strategy equilibrium. It is usually more difficult to adjust quality than to adjust price in manufacturing, but the quality of a financial product may be adjusted as easy as its price. We hope that this paper proves fruitful in generating further research into the competition between financial products.

A. Appendix: Proofs

This appendix contains the proofs of the main lemmas and propositions in the paper. See the Online Appendix for other proofs and robustness checks.

A.1 Proof of Proposition 1

Part (a): For $v_b \geq T^i$, the maximal surplus for a buyer to submit a limit order to exchange i is

$$BS^{i*}(v_b) = (v_b - P^{i*}(v_b) - f_m^i) \cdot \text{Prob}\left(v_s \leq \min\left\{P^{i*}(v_b) - f_t^i, \frac{1}{2}\right\}\right) = \frac{(v_b - T^i)^2}{2},$$

which depends only on the total fee T^i . Consequently, the liquidity maker chooses the exchange with the lowest total fee. For any v_b , the exchange with the lowest total fee earns a profit:

$$\Pi(T^i, v_b) = \begin{cases} T^i \cdot \text{Prob}(v_s \leq \min\{P^{i*}(v_b) - f_t^i, \frac{1}{2}\}) & \text{if } v_b \geq T^i \\ 0 & \text{if } v_b < T^i \end{cases}.$$

Accordingly, its total expected profit is given by

$$\begin{aligned} \Pi^{cont} &= \int_{v_b \geq T^i} \Pi(T^i, v_b) \cdot 2dv_b = 2T^i \cdot \int_{\max\{T^i, \frac{1}{2}\}}^1 \text{Prob}\left(v_s \leq \min\left\{P^{i*}(v_b) - f_t^i, \frac{1}{2}\right\}\right) dv_b \\ &= 2T^i \cdot \int_{\max\{T^i, \frac{1}{2}\}}^1 \text{Prob}\left(v_s \leq \frac{v_b - T^i}{2}\right) dv_b = 2T^i \cdot \int_{\max\{T^i, \frac{1}{2}\}}^1 (v_b - T^i) dv_b, \end{aligned}$$

which does not depend on how T^i is decomposed into f_m^i and f_t^i . Thus, only the total fee matters for the liquidity maker, the liquidity taker, and the exchanges.

Part (b) follows part (a). Since all traders go to the exchange with the lowest total fee, no operator can increase its profit by establishing multiple exchanges with a heterogeneous total fee. The monopoly operator's profit is

$$\Pi^{contM}(T) = 2T \cdot \int_{\max\{T, \frac{1}{2}\}}^1 (v_b - T) dv_b = \begin{cases} T \cdot (\frac{3}{4} - T) & \text{if } \frac{1}{2} \geq T \\ T \cdot (1 - T)^2 & \text{if } \frac{1}{2} < T \end{cases}.$$

The solutions are $T^{contM*} = \frac{3}{8}$ and $\Pi^{contM*} = \frac{9}{64}$.

Part (c) follows part (a) and Bertrand competition. The competition among multiple operators for total fees leads to a Bertrand equilibrium with zero total fee and zero profit. ■

A.2 Proof of Lemma 1

For traders to participate, the total fee should be less than the maximum gains from trade, so $f_m + f_t < 1$. When $f_m < 0$ and $f_t < 0$, exchanges cannot survive. Then $f_m \cdot f_t > 0$ requires $f_m > 0$ and $f_t > 0$. When $f_m > 0$, the liquidity maker must propose $P \leq 0$, and otherwise $P + f_m \geq 1 + f_m > 1$. When $f_t > 0$, the liquidity taker only accepts the limit order price at $P > 0$, otherwise the cum fee sell price $P - f_t$ is smaller than 0. So no trades occur when $f_m \cdot f_t > 0$. ■

A.3 Proof of Lemma 2

Part (a): Following Lemma 1, $f_m \geq 0$ implies $f_t < 0$ and $f_m + f_t \geq 0$ implies that $f_t \geq -f_m$. The other parts directly follow from the definitions of cum fee buy and sell prices.

Part (b): Takers' participation requires $p_s \in [0, \frac{1}{2}]$. $p_s = -(f_t - z) + P \in [0, \frac{1}{2}]$ and $P \in \mathbb{Z}$ requires $P = -z$, which leads to Equation (8). ■

A.4 Proof of Proposition 2

We denote p_b^i and p_s^i as exchange i 's cum fee buy price and cum fee sell price, respectively. Without loss of generality, we can assume that $p_s^1 \leq p_s^2 \leq \dots \leq p_s^K$. Then $p_b^1 \leq p_b^2 \leq \dots \leq p_b^K$. If not, suppose that exchange i has a lower cum fee buy price and a higher cum fee sell price than Exchange j . Then no liquidity maker chooses exchange j .

Next, we analyze liquidity makers' segmentation across K exchanges. A liquidity maker with valuation $v_b < p_b^1$ posts no limit order on any exchange. A liquidity maker with valuation $p_b^1 \leq v_b < p_b^2$ posts a limit order on exchange 1 only if

$$2 \cdot (v_b - p_b^2) \cdot p_s^2 \leq 2 \cdot (v_b - p_b^1) \cdot p_s^1, \quad (A1)$$

which is equivalent to $v_b \leq \frac{p_b^2 p_s^2 - p_b^1 p_s^1}{p_s^2 - p_s^1} \equiv \varphi^1$. A liquidity maker with valuation $v_b \in [p_b^1 \varphi^1]$ posts a limit order on exchange 1. Similarly, denote

$$\varphi^i = \begin{cases} p_b^1 & \text{for } i=0, \\ \frac{p_b^{i+1} p_s^{i+1} - p_b^i p_s^i}{p_s^{i+1} - p_s^i} & \text{for } 1 \leq i \leq K-1, \\ 1 & \text{for } i=K. \end{cases}$$

A liquidity maker with valuation $v_b \in [\varphi^{i-1}, \varphi^i]$ posts a limit order on exchange i for $1 \leq i \leq K$.

The monopoly operator chooses p_b^i and p_s^i on each exchange i to maximize

$$\begin{aligned} \Pi^M(K) &= 4 \sum_{i=1}^K (\varphi^i - \varphi^{i-1}) \cdot p_s^i \cdot (p_b^i - p_s^i) \\ \text{s.t. } \varphi^0 &\leq \varphi^1 \leq \dots \leq \varphi^K. \end{aligned}$$

The solutions are Equation (14). The profit of the monopoly operator is

$$\Pi^M(K) = 4 \sum_{i=1}^K \left(\frac{K+i+1}{2K+1} - \frac{K+i}{2K+1} \right) \cdot \frac{i}{2K+1} \cdot \left(\frac{1}{2} + \frac{i}{2(2K+1)} - \frac{i}{2K+1} \right) = \frac{2K \cdot (K+1)}{3(2K+1)^2}.$$

The liquidity maker's expected surplus is

$$BS^M(K) = 4 \sum_{i=1}^K (\varphi^i - \varphi^{i-1}) \cdot p_s^i \cdot \left(E(v_b | \varphi^{i-1} \leq v_b \leq \varphi^i) - p_b^i \right) = \frac{K \cdot (K+1)}{3(2K+1)^2}.$$

The liquidity taker's surplus is

$$SS^M(K) = 4 \sum_{i=1}^K (\varphi^i - \varphi^{i-1}) \cdot p_s^i \cdot (p_s^i - E(v_s | v_s \leq p_s^i)) = \frac{K \cdot (K+1)}{3(2K+1)^2}.$$

If a fixed cost c is involved in opening a new exchange, the monopoly operator keeps establishing the K^{th} exchange as long as $\Pi^M(K) - \Pi^M(K-1) = \frac{4K}{3(2K+1)^2(2K-1)^2} \geq c$. ■

A.5 Proof of Proposition 3

We prove Proposition 3 through contradiction. Without loss of generality, suppose that the pure-strategy equilibrium exists and that $\Pi^1 \geq \Pi^2$. Then a profitable deviating strategy always exists for exchange 2. Two cases are considered: (1) $\Pi^1 > 0$ and (2) $\Pi^1 = \Pi^2 = 0$.

- (a) There are two subcases: (1a) $\Pi^1 > \Pi^2 \geq 0$; (1b) $\Pi^1 = \Pi^2 > 0$.

(1a) Exchange 2 can set its fees such that

$$p_b^2 = p_b^1 - \varepsilon \text{ and } p_s^2 = p_s^1. \quad (\text{A2})$$

Liquidity makers and takers who originally chose exchange 1 will choose the new exchange 2 for any $\varepsilon > 0$. The profit of exchange 2 increases as long as ε is sufficiently small.

(1b) If exchanges 1 and 2 have the same cum fee sell price, they must have the same cum fee buy price to obtain equal profits. Deviation (A2) would nearly double exchange 2's profit. Now consider the case that they have heterogeneous cum fee sell prices but the same profit. Without loss of generality, assuming exchange 2 is the high-quality exchange. That is, $p_b^2 > p_b^1$ and $p_s^2 > p_s^1$. Then there is a cutoff $\varphi \in (\frac{1}{2}, 1)$, such that buyers with $v_b \geq \varphi > p_b^2$ choose exchange 2, while those with $\varphi > v_b \geq p_b^1$ choose exchange 1. Denote their identical profits as Π . By deviating from (p_b^2, p_s^2) to $(p_b^1 - \varepsilon, p_s^1)$, exchange 2 attracts all makers and takers who chose exchange 1 and maintains the liquidity makers who chose exchange 2. Next, we show that such deviation strictly improve exchange 2's profit.

Specifically, because the total fee for exchange 1 is $p_b^1 - p_s^1$, the original trading volume of exchange 1 is $\frac{\Pi}{p_b^1 - p_s^1}$. Therefore, exchange 1's original traders would add a profit of $\frac{\Pi}{p_b^1 - p_s^1} \cdot (p_b^1 - \varepsilon - p_s^1)$ to exchange 2 after its fee deviation. The liquidity makers' participation in exchange 2 is $\frac{\Pi}{2p_s^2 \cdot (p_b^2 - p_s^2)}$. All these liquidity makers would still choose the new exchange 2, because their valuation is higher than $\varphi > p_b^1 - \varepsilon$. The execution probability after deviation is $2p_s^1$. So liquidity makers and takers who originally would choose exchange 2 will provide a profit of $\frac{\Pi}{2p_s^2 \cdot (p_b^2 - p_s^2)} \cdot 2p_s^1 \cdot (p_b^1 - \varepsilon - p_s^1)$ to exchange 2 after deviation.

So exchange 2's profit after deviation is

$$\begin{aligned} & \frac{\Pi}{p_b^1 - p_s^1} \cdot (p_b^1 - \varepsilon - p_s^1) + \frac{\Pi}{2p_s^2 \cdot (p_b^2 - p_s^2)} \cdot 2p_s^1 \cdot (p_b^1 - \varepsilon - p_s^1) \\ &= \left[1 - \frac{\varepsilon}{p_b^1 - p_s^1} + \frac{p_s^1 \cdot (p_b^1 - \varepsilon - p_s^1)}{p_s^2 \cdot (p_b^2 - p_s^2)} \right] \cdot \Pi \\ &> \Pi \text{ if } \varepsilon < \frac{p_s^1 \cdot (p_b^1 - p_s^1)^2}{p_s^1 \cdot (p_b^1 - p_s^1) + p_s^2 \cdot (p_b^2 - p_s^2)}. \end{aligned}$$

- (b) Both exchanges earn zero profits if (2a) no trading or (2b) $p_b^i = p_s^i \in (0, 1)$, $i = 1, 2$,

(2a) Exchange 2 can earn positive profits by setting $0 < p_s^2 < p_b^2 < 1$.

(2b) Denote $p_b^1 = p_s^1 = \gamma$. Two subcases are (2bi) $0 < \gamma < \frac{1}{2}$ and (2bii) $\frac{1}{2} \leq \gamma < 1$.

(2bi) Exchange 2 can set its fees such that

$$p_b^2 = \gamma + \varepsilon \text{ and } p_s^2 = \gamma + \mu \cdot \varepsilon, \quad (\text{A3})$$

where $\varepsilon > 0$ and $0 < \mu < 1$. The liquidity maker's segmentation cutoff is

$$\varphi = \frac{p_b^2 \cdot p_s^2 - p_b^1 \cdot p_s^1}{p_s^2 - p_s^1} = \gamma \left(1 + \frac{1}{\mu} \right) + \varepsilon.$$

Note that φ decreases with μ . For μ sufficiently close to 1 and ε sufficiently close to zero, $\lim_{\mu \rightarrow 1} \varphi = 2\gamma + \varepsilon < 1$, where the inequality follows from $\gamma < \frac{1}{2}$. A liquidity maker with

valuation $v_b \in (\varphi, 1]$ chooses exchange 2, and trades with a liquidity taker with valuation $v_s \in (0, \gamma + \mu \cdot \varepsilon]$. Exchange 2 earns a strictly positive profit $p_b^2 - p_s^2 = (1 - \mu) \cdot \varepsilon$ from such liquidity makers and liquidity takers.

(2bii) Exchange 2 can set its fees such that

$$p_b^2 = \gamma - \mu \cdot \varepsilon \text{ and } p_s^2 = \gamma - \varepsilon, \quad (\text{A4})$$

where $\varepsilon > 0$ and $0 < \mu < 1$.

When $\gamma > \frac{1}{2}$, for sufficiently small ε , $p_s^2 = \gamma - \varepsilon \geq \frac{1}{2}$ and $p_b^2 < p_b^1$. Liquidity makers with valuation above $\gamma - \mu \cdot \varepsilon$ then choose exchange 2, and trade with any liquidity taker. Exchange 2 earns a profit of $p_b^2 - p_s^2 = (1 - \mu) \cdot \varepsilon > 0$ per trade from these traders.

When $\gamma = \frac{1}{2}$, the liquidity maker's segmentation cutoff is

$$\varphi = \frac{p_b^2 \cdot p_s^2 - p_b^1 \cdot p_s^1}{p_s^2 - p_s^1} = \frac{1}{2} + \mu \cdot \left(\frac{1}{2} - \varepsilon \right).$$

For sufficiently small ε , $\varphi > \frac{1}{2}$. Liquidity makers with valuation $\frac{1}{2} \leq v_b \leq \varphi$ choose exchange 2. Exchange 2 earns a strictly positive profit, as $p_b^2 - p_s^2 = (1 - \mu) \cdot \varepsilon > 0$. ■

A.6 Proof of Proposition 4

Part (a): Our game satisfies sufficient conditions for the existence of a symmetric mixed-strategy equilibrium, as specified in theorem 6* of Dasgupta and Maskin (1986). We verify such sufficient conditions in the Online Appendix.

Part (b): No mixed-strategy equilibrium should entail negative profit. We then prove the nonexistence of the mixed-strategy equilibrium with zero profit by contradiction.

1. If the mixed-strategy equilibrium involves zero profit, the probability that the effective buy price is greater than 0.5 must be zero, that is, $\text{prob}(x < p_b < y) = 0$ for any $x > 0.5$. If not, one exchange can set $\tilde{p}_b = \frac{0.5+x}{2}$ and $\tilde{p}_s = 0.5$ and attract liquidity makers with valuation $v_b \in [\frac{0.5+x}{2}, x]$ with positive probability and make positive profits.
2. If the mixed-strategy equilibrium involves zero profit, the probability that the effective buy price is smaller than 0.5 must be zero, that is, $\text{prob}(x < p_b < y) = 0$ for any $y < 0.5$. If not, suppose that exchange 1 set $\text{prob}(x < p_b < y) > 0$ for some $y < 0.5$. Then we show that exchange 2 can set $\tilde{p}_b = 0.5$ and $\tilde{p}_s = 0.5 - \varepsilon$ ($\varepsilon > 0$) and attract positive volume. To see that, when liquidity maker chooses exchange 1 under cum fee buy price $p_b = \lambda < 0.5$, the maximum surplus is $2(\varphi - \lambda)\lambda$, because $p_b \geq p_s$. The proof is equivalent to showing that, for sufficiently small ε , we can find $\varphi < 1$, such that

$$(\varphi - 0.5)(0.5 - \varepsilon) = (\varphi - \lambda)\lambda$$

$$\varphi = \frac{0.25 - 0.5\varepsilon - \lambda^2}{0.5 - \varepsilon - \lambda}$$

$$\varphi < 1 \iff \varepsilon < 2(0.5 - \lambda)^2.$$

Therefore, $\tilde{p}_b = 0.5$, $\tilde{p}_s = 0.5 - \varepsilon$ attracts any liquidity maker with valuation $v_b \in [\varphi, 1]$ when exchange 1 quotes a cum fee buy price of less than 0.5 with positive probability.

3. If the mixed-strategy equilibrium involves zero profit, the probability that the effective buy price is equal to 0.5 must be zero. If not, one exchange can use strategy (A4) to obtain positive profit with positive probability.

Combining 1–3, the mixed-strategy equilibrium must have positive profit. ■

A.7 Proof of Lemma 3

For any v_b , the buyer's expected surplus under (p_b, p_s) with $p_b \geq p_s$ is

$$BS(v_b) \equiv (v_b - p_b) \cdot 2p_s,$$

which is maximized when $p_b = p_s = \frac{v_b}{2}$. ■

A.8 Proof of Lemma 4

First, $\frac{a^2+b^2}{a+b} - \frac{a+b}{2} = \frac{a^2-2ab+b^2}{2(a+b)} = \frac{(a-b)^2}{2(a+b)} > 0$. Thus $T = p_b - p_s > 0$.

Next, we show this new fee structure attracts positive volume.

1. Lemma 3 shows that liquidity makers strictly prefer $p_b = p_s = \frac{v_b}{2}$. When $P_s \cap (a, b) = \emptyset$, liquidity makers at $v_b \in (2a, 2b)$ cannot achieve their maximum surplus under breakeven constraints. Because $BS(v_b)$ decreases in p_b , the maximization of $BS(v_b)$ requires $p_b = p_s$. Because $BS(v_b) = (v_b - p_s) \cdot 2p_s$ is quadratic in p_s , liquidity makers' surplus must be lower than either the one under fee structure (a, a) or the one under fee structure (b, b) .
2. Because $a < \frac{a+b}{2} < b$, the new free structure attracts liquidity makers with valuation higher than $\varphi_a \equiv \frac{p_b p_s - a^2}{p_s - a}$ and lower than $\varphi_b \equiv \frac{b^2 - p_b p_s}{b - p_s}$. To show that the new structure attracts positive volume, we need to show that $2a < \varphi_a < \varphi_b < 2b$.
 - a. $\varphi_a = \frac{p_b p_s - a^2}{p_s - a} > \frac{(p_s)^2 - a^2}{p_s - a} = p_s + a > 2a$.
 - b. $\varphi_b = \frac{b^2 - p_b p_s}{b - p_s} < \frac{b^2 - (p_s)^2}{b - p_s} = b + p_s < 2b$.
 - c. As $p_s - a = b - p_s$ $\varphi_a < \varphi_b \Leftrightarrow p_b p_s - a^2 < b^2 - p_b p_s \Leftrightarrow p_b < \frac{a^2 + b^2}{a + b}$.

Also, it is easy to verify that $\varphi_a > p_b$. Hence, liquidity makers with $v_b \in (\varphi_a, \varphi_b) \subset (2a, 2b)$ will participate and prefer the new exchange with our constructed cum fee buy and sell prices over all other exchanges. ■

A.9 Proof of Proposition 5

(Sufficiency) No operator has profitable deviation, because the fee structures from its competing operators span all liquidity makers' ideal price levels with breakeven constraints.

(Necessity) First, in equilibrium, total fees at all active exchanges must be zero. Otherwise, an exchange with positive profit would attract new entrants who can mimic its fee structure and share part of its positive profit. Therefore, $p^{i,k} \equiv p_b^{i,k} = p_s^{i,k}$ for all active exchanges (i, k) . Second, let I be the set of operators and K be the set of exchanges. Denote the collection of $p^{i,k}$ that does not belong to operator i as $\mathcal{P}^{-i}$. Suppose $\exists x \in [\frac{1}{4}, \frac{1}{2}]$ and $\exists \varepsilon > 0$, such that $\mathcal{P}^{-i} \cap (x - \varepsilon, x + \varepsilon) = \emptyset$. Then let $a = \max\{\frac{1}{4}, x - \varepsilon\}$ and $b = \min\{\frac{1}{2}, x + \varepsilon\}$. Operator i can set only one exchange in (a, b) with cum fee buy and sell prices outlined in Lemma 4 and obtain positive profits. ■

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